

EVALUATION REPORT

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Week 3: Machine Learning for AI

Introduction

This project applied a Machine Learning classification model to predict passenger survival using the Titanic dataset. The objective was to understand the machine learning workflow, including data preprocessing, model training, evaluation, and interpretation of results.

Type of Model Used

A Logistic Regression model was used for this project. Logistic Regression is a supervised machine learning algorithm commonly used for classification tasks where the target variable has two possible outcomes. In this case, the model was used to predict whether a passenger survived or did not survive the Titanic disaster.

Reason for Model Selection

Logistic Regression was selected because the target variable (Survived) is binary in nature. The model is simple, efficient, easy to interpret, and suitable for introductory classification problems. It also provides a good foundation for understanding supervised learning concepts before exploring more advanced machine learning algorithms.

Model Performance and Evaluation Results

The dataset was divided into training and testing sets, and the Logistic Regression model was trained using passenger features including passenger class, gender, age, and fare.

The model achieved an accuracy score of 80.45%, indicating that it correctly classified approximately 80% of passenger survival outcomes.

Additional evaluation metrics showed strong overall performance. For passengers who did not survive, the model achieved a precision of 82% and a recall of 86%. For passengers who survived, the model achieved a precision of 78% and a recall of 73%. These results demonstrate that the model was effective in identifying survival patterns within the dataset while maintaining balanced predictive performance.

Challenges Encountered

One challenge encountered during implementation was preparing the dataset for machine learning by converting categorical variables into numerical formats suitable for model training. Another challenge was understanding the meaning and interpretation of evaluation metrics such as precision, recall, and accuracy. Interpreting model outputs and feature contributions also required careful analysis to draw meaningful conclusions.

Key Insights Learned

This project demonstrated the practical application of supervised machine learning for classification tasks. It reinforced the importance of data preprocessing and feature selection in building effective models. The Logistic Regression model achieved an accuracy of 80.45%, indicating strong predictive performance for a beginner classification model.

Feature importance analysis revealed that gender was the most influential factor affecting survival predictions, followed by passenger class. This finding supports the patterns previously identified during Exploratory Data Analysis, where female passengers and first-class passengers showed higher survival rates. Age and fare contributed less to the model's predictions.

Overall, the project provided valuable experience in data preparation, model training, evaluation, and interpretation, while demonstrating how machine learning can be used to uncover meaningful patterns and support data-driven decision-making.

Conclusion

The Logistic Regression model successfully predicted Titanic passenger survival with an accuracy of 80.45%. Through this project, I gained hands-on experience with data preprocessing, model training, evaluation, and interpretation, strengthening my understanding of machine learning as a core component of Artificial Intelligence systems.